

ECON426 Applied Econometrics

Introduction to

LASSO Estimator

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Why do we even need LASSO?

Why do we even need LASSO?

- ▶ Too many regressors $\Rightarrow$ high variance and unstable estimates
- ▶ Multicollinearity $\Rightarrow$ leads to noisy coefficients
- ▶ High-dimensional settings $\Rightarrow$ weak interpretability
- ▶ Prediction and inference should not be the same.

Why does OLS fail when $p > n$?

$$Y = X\beta + \varepsilon, \quad X \in \mathbb{R}^{n \times p}$$

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'Y$$

If $p > n$, then

$$\text{rank}(X) \leq n < p$$

so the columns of X are linearly dependent. Therefore,

$$X'X \text{ is singular} \quad \Rightarrow \quad (X'X)^{-1} \text{ does not exist}$$

Interpretation

When the number of candidate regressors exceeds the number of observations, OLS cannot uniquely estimate the coefficient vector.

High-Dimensionality problem: Growth Example

Suppose we want to explain growth using:

- ▶ initial income
- ▶ schooling
- ▶ investment
- ▶ population growth
- ▶ inflation

But we only observe 3 countries.

$$X = \begin{bmatrix} 1 & \text{income}_1 & \text{schooling}_1 & \text{investment}_1 & \text{popgrowth}_1 & \text{inflation}_1 \\ 1 & \text{income}_2 & \text{schooling}_2 & \text{investment}_2 & \text{popgrowth}_2 & \text{inflation}_2 \\ 1 & \text{income}_3 & \text{schooling}_3 & \text{investment}_3 & \text{popgrowth}_3 & \text{inflation}_3 \end{bmatrix}$$

$$X \in \mathbb{R}^{3 \times 6} \Rightarrow n = 3, p = 6, p > n$$

High-dimensionality problem: Labor Market example

Suppose we want to explain wages using:

- ▶ education
- ▶ experience
- ▶ tenure
- ▶ gender
- ▶ region dummies

But we only observe 4 individuals.

$$X \in \mathbb{R}^{4 \times 6} \quad \Rightarrow \quad n = 4, \quad p = 6, \quad p > n$$

$$X = \begin{bmatrix} 1 & \text{educ}_1 & \text{exp}_1 & \text{tenure}_1 & \text{occ}_1 & \text{region}_1 \\ 1 & \text{educ}_2 & \text{exp}_2 & \text{tenure}_2 & \text{occ}_2 & \text{region}_2 \\ 1 & \text{educ}_3 & \text{exp}_3 & \text{tenure}_3 & \text{occ}_3 & \text{region}_3 \\ 1 & \text{educ}_4 & \text{exp}_4 & \text{tenure}_4 & \text{occ}_4 & \text{region}_4 \end{bmatrix}$$

Ridge regression replaces OLS:

$$\hat{\beta}_{\text{ridge}} = \arg \min_{\beta} \left\{ \sum_{i=1}^n (y_i - x_i' \beta)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}$$

Let

$$L(\beta) = \|Y - X\beta\|_2^2 + \lambda \beta' \beta$$

$$\frac{\partial L(\beta)}{\partial \beta} = -2X'Y + 2X'X\beta + 2\lambda\beta$$

Setting the derivative equal to zero:

$$(X'X + \lambda I)\beta = X'Y$$

So the ridge estimator is:

$$\boxed{\hat{\beta}_{\text{ridge}} = (X'X + \lambda I)^{-1} X'Y}$$

Adding λI makes the problem well-defined and stabilizes estimation.

The LASSO Objective Function

LASSO solves:

$$\min_{\beta} \left\{ \underbrace{\sum_{i=1}^n (y_i - x_i' \beta)^2}_{\text{Residual Sum of Squares (RSS)}} + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

- First term: fit to the data (RSS)
- Second term: penalty on coefficient size
- λ : controls sparsity

Statistical properties of LASSO

LASSO has different goals than OLS:

- ▶ Not unbiased:

$$E[\hat{\beta}^{\text{lasso}}] \neq \beta$$

- ▶ Introduces bias intentionally (shrinkage)

Correctly identifies non-zero coefficients

- ▶ Consistent for prediction:

$$\sum_{i=1}^n (x_i'(\hat{\beta} - \beta))^2 \rightarrow 0$$

Key idea

LASSO trades bias for lower variance and variable selection.

What does Least Absolute Shrinkage and Selection Operator do?

- ▶ It adds an ℓ_1 penalty to the least-squares objective.
- ▶ Small coefficients are shrunk more aggressively.
- ▶ Some coefficients become exactly zero.
- ▶ So LASSO is both a shrinkage method and a variable selection method.

Interpretation

Large $\lambda \Rightarrow$ more shrinkage, fewer selected variables.

Small $\lambda \Rightarrow$ less shrinkage, more selected variables.

LASSO as an optimization problem

LASSO is defined as:

$$\hat{\beta}^{\text{lasso}} = \arg \min_{\beta} \left\{ \sum_{i=1}^n (y_i - x_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

- ▶ No closed-form solution (unlike OLS or Ridge)
- ▶ The objective function is convex
- ▶ The solution is obtained using numerical optimization algorithms

Analogy

Similar to Maximum Likelihood Estimation:

- ▶ define an objective function
- ▶ solve using optimization methods

How is LASSO computed?

Since there is no closed-form solution, LASSO is solved numerically.

Common algorithms:

- ▶ Coordinate Descent
- ▶ Gradient-based methods

Idea of coordinate descent:

- ▶ Update one coefficient at a time
- ▶ Apply soft-thresholding
- ▶ Repeat until convergence

Modern software (Stata, R, Python) implements these efficiently

Solving LASSO: idea of coordinate descent

LASSO solves:

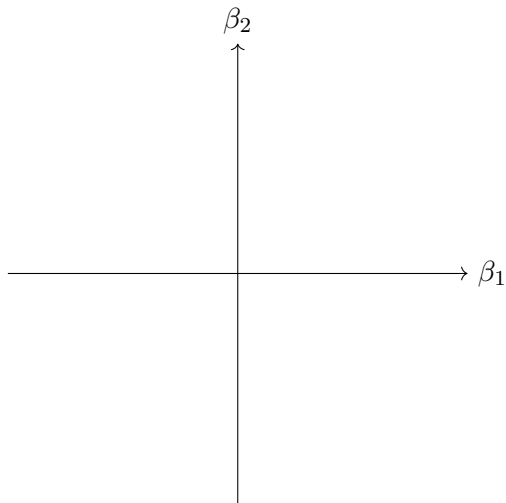
$$\hat{\beta} = \arg \min_{\beta} \left\{ \sum_{i=1}^n (y_i - x_i' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}$$

Instead of solving for all coefficients at once:

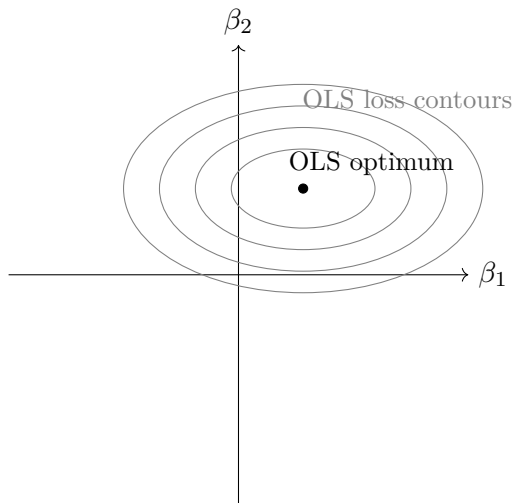
- ▶ Fix all coefficients except β_j
- ▶ Minimize with respect to β_j
- ▶ Repeat for all $j = 1, \dots, p$

Break a high-dimensional problem into simple one-dimensional problems

Geometry of Ridge vs LASSO

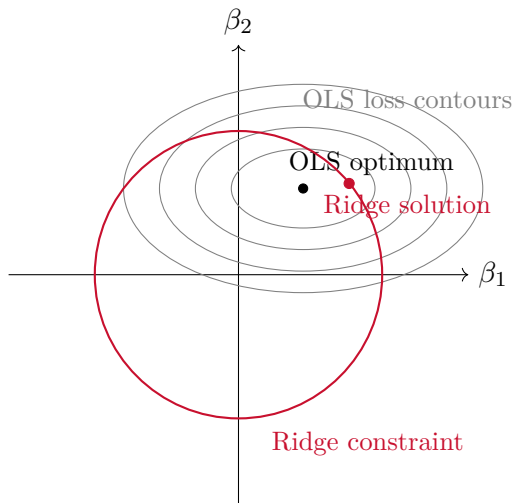


Geometry of Ridge vs LASSO



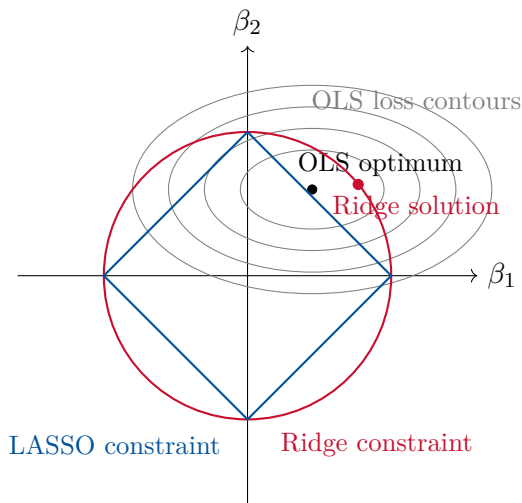
The ellipses show combinations of coefficients with the same residual sum of squares.

Geometry of Ridge vs LASSO



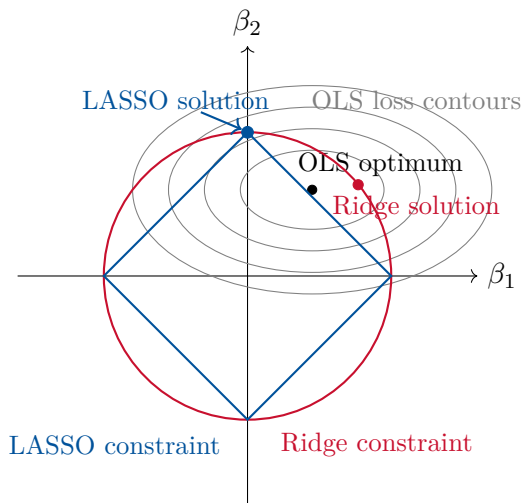
Ridge adds an ℓ_2 constraint: coefficients are shrunk, but usually not set exactly to zero.

Geometry of Ridge vs LASSO



LASSO uses an ℓ_1 constraint, which has corners instead of a smooth boundary.

Geometry of Ridge vs LASSO



Corners make zero coefficients likely $\Rightarrow$ LASSO performs variable selection.

Why the Diamond Matters: Geometry

- ▶ Ridge uses an ℓ_2 ball: smooth, circular.
- ▶ LASSO uses an ℓ_1 ball: diamond-shaped.
- ▶ The least-squares contours are ellipses.
- ▶ The first contact point with the diamond often occurs at a corner.
- ▶ Corners correspond to coefficients equal to zero.

Visual message

$$|\beta_1| + |\beta_2| \leq c$$

diamond $\Rightarrow$ sparsity

This is why LASSO selects variables while ridge usually only shrinks them.

Method	Penalty	Main strength	Main weakness
OLS	none	Unbiased under classical assumptions	Fails or becomes unstable in high dimensions
Ridge	ℓ_2	Stabilizes coefficients, handles collinearity	Usually keeps all variables in the model
LASSO	ℓ_1	Shrinkage + variable selection	Introduces bias; selection can be unstable with correlated regressors

Applied example: returns to education

We want to estimate the causal effect of education on wages:

$$\text{wage}_i = \beta_1 \cdot \text{educ}_i + \mathbf{X}_i' \gamma + \varepsilon_i$$

Dataset:

- ▶ Individuals from labor survey
- ▶ Outcome: wages
- ▶ Treatment: years of education
- ▶ Controls: demographics, experience, region, occupation, etc.

Based on Belloni et al. (2014)

Potentially hundreds of control variables

LASSO selects relevant controls

Using LASSO to select controls:

- ▶ Start with 150 candidate variables
- ▶ LASSO selects a small subset

Selected variables

- ▶ experience
- ▶ experience²
- ▶ gender
- ▶ region dummies
- ▶ occupation dummies

Most variables are dropped (coefficients = 0)

Results: effect of education

Method	$\hat{\beta}_{\text{educ}}$	Std. Error
Naive OLS	0.02	0.015
OLS with many controls	0.05	0.030
Post-LASSO / Double LASSO	0.08	0.012

Interpretation

- ▶ Naive OLS underestimates effect
- ▶ Too many controls $\rightarrow$ noisy estimates
- ▶ LASSO selects relevant controls and improves inference

Applied case 2: portfolio allocation in high dimensions

In portfolio optimization, key quantities depend on the inverse covariance matrix:

$$\Theta = \Sigma^{-1}$$

Examples:

$$w_{\text{GMV}} = \frac{\Sigma^{-1} \mathbf{1}}{\mathbf{1}' \Sigma^{-1} \mathbf{1}}$$

where w_{GMV} is the global minimum variance portfolio.

- ▶ p = number of assets
- ▶ n = number of time observations

If $p > n$, the sample covariance matrix is singular, so direct inversion is problematic.

Based on Callot, Caner, Onder, and Ulasan (2019)

Nodewise regression: estimating Σ^{-1} with LASSO

For each asset $j = 1, \dots, p$, regress its demeaned return on all the others:

$$r_{t,j}^* = (r_{t,-j}^*)' \gamma_j + \eta_{t,j}$$

Estimate γ_j by LASSO:

$$\hat{\gamma}_j = \arg \min_{\gamma \in \mathbb{R}^{p-1}} \left(\frac{1}{n} \sum_{t=1}^n (r_{t,j}^* - r_{t,-j}^* \gamma)^2 + 2\lambda_j \sum_{k \neq j} |\gamma_k| \right)$$

Repeat this for all $j = 1, \dots, p$, then combine the estimated rows to build an estimate of the precision matrix.

Idea: estimate the inverse covariance matrix directly, without inverting a singular sample covariance matrix.

Real high-dimensional example from the paper

Empirical application in the paper:

$n = 293$ monthly observations, $p = 304$ assets

So:

$$R \in \mathbb{R}^{293 \times 304}, \quad p > n$$

The authors compare Nodewise LASSO, POET, and Ledoit–Wolf methods.

Main finding:

- ▶ Nodewise regression-based portfolios perform well out of sample
- ▶ especially in terms of Sharpe ratio, variance, and turnover on monthly data

Illustrative output: monthly portfolio performance

Method	Return	Variance	Sharpe Ratio
POET	0.02499	0.01953	0.1788
Nodewise	0.02644	0.01580	0.2104
Ledoit–Wolf	0.07140	0.18421	0.1664

Nodewise delivers a higher Sharpe ratio with lower variance in this monthly out-of-sample exercise.

LASSO in Stata: Main Commands

► Prediction / model selection:

```
lasso linear y controls  
elasticnet linear y controls  
sqrtlasso y controls
```

► Inference:

```
dsregress y d, controls(controls)  
poregress y d, controls(controls)  
xporegress y d, controls(controls)
```

Interpretation

Use lasso for prediction; use dsregress/poregress/xporegress for inference.